Boundary-Aware Multiscale Learning Perception for Remote Sensing Image Segmentation

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***Abstract*— For remote sensing image segmentation, the boundaries of objects are difficult to distinguish, which is ignored by most methods. Therefore, it is challenging how to excavate and recover the boundaries of objects accurately. In this article, we propose a boundary-aware multiscale network (BMNet) to solve this problem. The key components of BMNet include the scale attention module (SAM) and boundary guidance module (BGM). Specifically, the SAM is proposed to guide the refinement of multiscale features in a context-aware way. It enhances the discriminability of multiscale features by establishing contextual dependencies, which enables the refinement of the prediction of objects. Then, the BGM is proposed to enable networks to distinguish the boundary of objects. It uses manifold information of features to generate BG maps and forces the network to focus more on the boundary of objects. The effectiveness of the proposed BMNet is demonstrated on two public remote sensing datasets: ISPRS 2-D semantic labeling Potsdam dataset and Vaihingen dataset, where BMNet achieves better segmentation than prevalent methods. Finally, the experimental results indicate that BMNet can produce sharper boundaries of objects to reconstruct more detailed segmentation results.**

***Index Terms*— Boundary attention guidance, learning perception, multiscale, remote sensing, semantic segmentation.**

Manuscript received 14 April 2023; revised 21 June 2023 and 20 July 2023; accepted 26 July 2023. Date of publication 1 August 2023; date of current version 18 August 2023. This work was supported in part by the Key Scientific Technological Innovation Research Project by Ministry of Education; in part by the National Natural Science Foundation of China Innovation Research Group Fund under Grant 61621005; in part by the State Key Program and the Foundation for Innovative Research Groups of the National Natural Science Foundation of China under Grant 61836009; in part by the Major Research Plan of the National Natural Science Foundation of China under Grant 91438201, Grant 91438103, and Grant 91838303; in part by the National Natural Science Foundation of China under Grant U1701267, Grant 62076192, Grant 62006177, Grant 61902298, Grant 61573267, Grant 61906150, and Grant 62276199; in part by the 111 Project; in part by the Program for Cheung Kong Scholars and Innovative Research Team in University under Grant IRT 15R53; in part by the Science and Technology (ST) Innovation Project from the Chinese Ministry of Education; in part by the Key Research and Development Program in Shaanxi Province of China under Grant 2019ZDLGY03-06; in part by the National Science Basic Research Plan in Shaanxi Province of China under Grant 2019JQ-659 and Grant 2022JQ-607; in part by the China Postdoctoral Fund under Grant 2022T150506; in part by the Scientific Research Project of Education Department in Shaanxi Province of China under Grant 20JY023; in part by the Fundamental Research Funds for the Central Universities under Grant XJS201901, Grant XJS201903, Grant JBF201905, and Grant JB211908; and in part by the Chinese Association of Artificial Intelligence (CAAI)-Huawei MindSpore Open Fund. *(Corresponding author: Licheng Jiao.)*

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Digital Object Identifier 10.1109/TGRS.2023.3300706

I. Introduction

SEMANTIC segmentation aims to assign a category label to each pixel in images, which is of great significance for image understanding and analysis [1], [2], [3]. Therefore, semantic segmentation is of great value for remote sensing in many applications, such as automated driving [4], urban planning [5], and intelligent agriculture [6]. Over the years, semantic segmentation has always been a widely researched field. Traditional methods [7], [8], [9] have demands for handcrafted features, which cannot effectively represent the semantic information of remote sensing images, and result in bad segmentation.

Recently, a large number of novel segmentation methods have emerged, especially deep learning (DL) methods. Different from traditional methods, DL methods [10], [11], [12] have achieved great success in remote sensing image segmentation, thanks to their powerful ability to extract features. Unfortunately, these methods focus on aggregating richer contextual information and lead to a critical loss of boundary information during continuous downsampling, which makes it difficult to distinguish the boundaries of objects and obtain precise segmentation results.

To reduce the loss of boundary information and obtain detailed segmentation results, many methods have been proposed to solve this issue. Li et al. [13] and Heidler et al. [14] used hierarchical attention to refine boundaries in the process of aggregating multiscale features. Novel loss functions are proposed to constrain the learning process of boundaries in [15] and [16]. The boundary detection branch is constructed in [17] to obtain boundary information in a supervised manner. Wei et al. [18] took advantage of the proposed novel convolution to achieve high-precision boundary regression. Zhu et al. [19] and Yuan et al. [20] deformed the boundary iteratively by updating offsets in a learnable manner. Zhou et al. [21] generate a directional relationship between pixels to guide fine segmentation of boundaries.

However, the above methods still cannot achieve satisfactory segmentations. Due to the scale variation [22] of objects and the noise [23] from shadows, and occlusions in remote sensing images, the discriminability of features is greatly influenced. These methods alleviate the effect of scale variations to some extent in a multiscale way, but they cannot effectively remove the noise to provide enough discriminative features. Therefore,

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it is challenging for remote sensing how to obtain multiscale contextual information to enhance the discrimination of features. In addition, prevalent methods obtain semantic features with convolution layers, which cannot explicitly extract boundary information from remote sensing images [24]. This disadvantage will result in an irreversible loss of boundary information during continuous convolutions, and the barren boundary information that remains in features will make it difficult in distinguishing the boundary of objects. Therefore, how to explicitly extract effective boundary information is an urgent problem.

Based on the above considerations, a boundary-aware multiscale network (BMNet) is proposed to achieve detailed segmentation of remote sensing images. First, with the introduction of the self-attention mechanism, the proposed scale attention module (SAM) constructs long-range contextual dependencies in a multiscale way to effectively eliminate the noise in remote sensing images. The contextual dependencies increase the diversity of features from a spatial and channel perspective, which has the advantage to provide more discriminative features during representing features of objects. Second, the proposed boundary guidance module (BGM) extracts the explicit boundary information to guide the refinement of the segmentation near the boundary. Explicit boundary information can effectively relieve the loss of boundary information and makes the network focus on regions near the boundary. Finally, the visual and quantitative experiments are conducted on remote sensing datasets, showing that the BMNet effectively improves semantic segmentation performance.

The main contributions of this article can be given as follows.

1) A BMNet is constructed to refine the segmentation of remote sensing images with the guidance of multiscale contextual information and explicit boundary information.
2) A novel SAM is proposed that can capture long-range contextual information in a multiscale way to enhance the discriminability of features. The SAM constructs multiscale contextual dependencies with a self-attention mechanism to eliminate the noise of features, thereby making features more discriminative.
3) To explicitly extract boundary information, a novel BGM is proposed. The BGM uses high-dimensional manifold information of objects to adaptively generate a BG map in an explicit manner, which guides networks to focus on the recovery of features near the boundary.
4) The visual and quantitative experiments are conducted on two remote sensing datasets to validate the effectiveness of the BMNet, and experiments show that the BMNet achieves better segmentation than popular methods.

The remainder of this article is organized as follows. Section II introduces related works about this article. Section III introduces our BMNet architecture and its components in detail, including the SAM and BGM. Section IV presents the experimental setup and the detailed experimental analysis. Finally, the conclusions of this article are drawn in Section V.

II. Related Works

In this section, the related methods are reviewed, which are widely researched. These methods are divided into three categories to be presented: remote sensing image semantic segmentation, attention mechanisms, and boundary detection.

A. Remote Sensing Image Semantic Segmentation

Semantic segmentation, as one of the fundamental tasks in computer vision [25], [26], [27], [28], has been widely studied in remote sensing. Traditional methods tend to use region split–merge (RSM) methods and active contour models (ACMs). The RSM methods usually oversegment pixels in the image into small regions called superpixels, and then merge similar superpixels to obtain the final segmentation, such as K-means [29], meanshift [30], watershed [31], simple linear iterative clustering (SLIC) [9], and normalized cuts [7]. The ACM models, such as [32], [33], and [34], guide the motion of the boundary curve by the internal spline energy and external constraint energy to generate the final segmentation. However, traditional methods all depend excessively on handcrafted features, which cannot represent features of objects well.

Compared with traditional methods, DL methods have achieved successful development in remote sensing image segmentation, which avoids the difficulties of handcrafted features and can adaptively extract more robust features. Peng et al. [35] constructed a fully convolutional network (FCN) [36] which performs the segmentation task in an end-to-end way. The U-shaped structure [37] is used in [38] to aggregate multiscale features with skip connections. Li et al. [12] and Liu et al. [39] designed networks based on dilation convolutions [40] to aggregate the contextual information. The pyramid pooling module (PPM) [41] is constructed in [42], which obtains multiscale features with several pooling layers. Sun et al. [43] enhanced the representation of features by maintaining high-resolution features in the encoder–decoder structures. Zhang et al. [44] and Xiao et al. [45] constructed the encoder structures with transformer blocks [46]. The above methods are all prevalent and achieved great success in the field of remote sensing image segmentation.

B. Attention Mechanisms

Inspired by human perception, the attention mechanisms can guide networks to find salient regions of scenes [47], which can greatly improve the representation of features. In [48], a channel attention module is proposed to model relationships between channels. Woo et al. [49] extended this idea to the spatial dimension and refined features with both the channel and spatial attentions. Qin et al. [50] and Wang et al. [51] developed different variants of the channel and spatial attentions. A novel nonlocal attention [52] is proposed to construct the spatial contextual dependencies, which can efficiently aggregate spatial contextual information. Huang et al. [53] optimized the computational efficiency of contextual information with a criss-cross attention module. Fu et al. [54] extended this idea to the channel dimension

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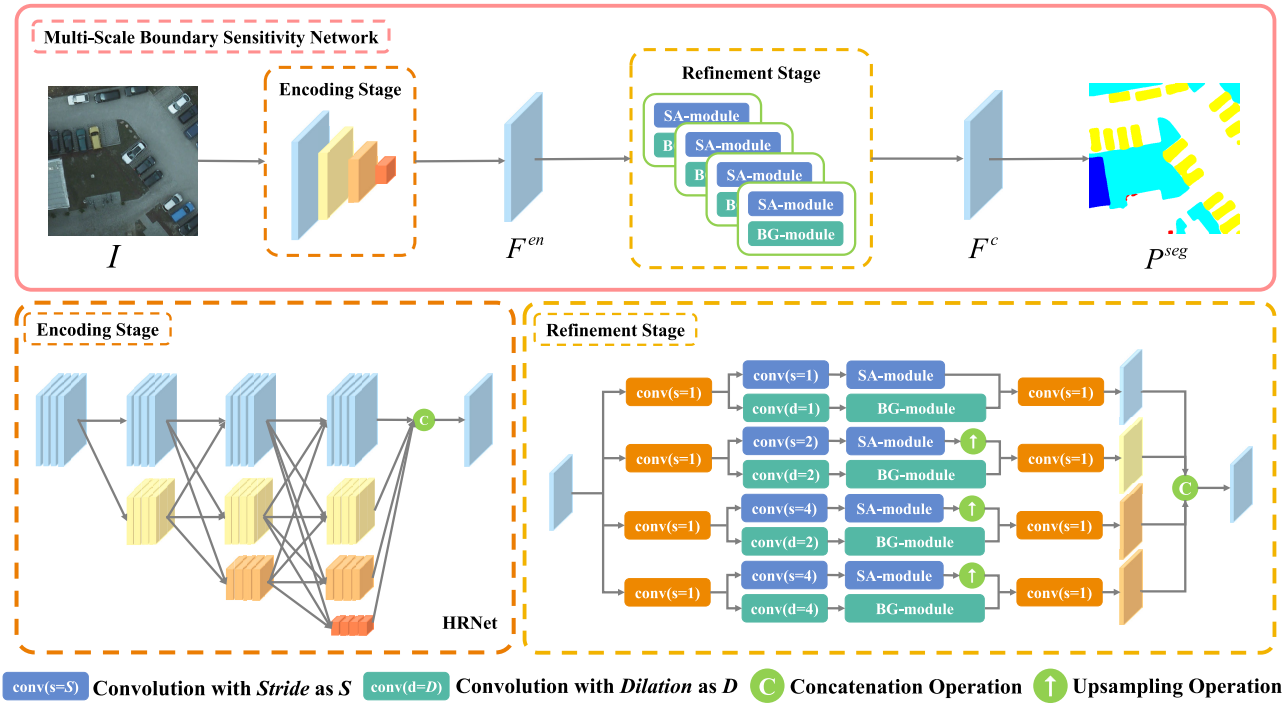


Fig. 1. Overview of BMNet. First, the HRNet is used as an encoder to extract multiscale features in the encoding stage. Then, in the refinement stage, features are refined by the SAM and BGM in a multiscale manner. Different scale factors in branches control the downsampling rate or dilation rate of convolutions in the proposed modules to ensure the scale diversity of features. Finally, multiscale features are concatenated and then fed into the classifier to generate the final segmentation.

and modeled the semantic dependencies both in the spatial and channel dimensions. Based on these attention methods, Li et al. [55] and He et al. [56] proposed different types of nonlocal mechanisms to aggregate various forms of contextual dependencies. The above research works focus on effective access to contextual dependencies to refine the representation of features for semantic segmentation.

C. Boundary Detection

Boundary detection has always played an important role in computer vision, and boundary maps generated by it provide vital information for various image processing tasks [57]. Early methods detected low-level boundaries mainly by first-order or second-order derivatives of an image, such as Sobel [58], Canny [59], and LoG [60]. Recently, with the introduction of convolutional neural networks (CNNs), boundary detection methods enable detection of more complex boundaries in an adaptive manner. Holistically-nested edge detection (HED) [61] learned rich hierarchical boundary representations by means of CNNs in a supervised manner. Deep category-aware semantic edge detection (CASENet) [62] associated each edge pixel with more than one class to represent junctions belonging to two or more semantic classes. Pixel difference network (PiDiNet) [63] and BSNet [24] designed a novel dynamic convolution operator for boundary detection. Edge detection Transformer (EDTER) [64] excavates the boundary information with transformers. With the assistance of boundary information, the details of the segmentation can be greatly refined.

III. Boundary-Aware Multiscale Network

In this section, BMNet is proposed to refine feature representations near the boundary. First, the overview of BMNet is introduced. Second, the SAM is proposed to enhance the discriminatory of features. Third, the BGM and its key component pixel cosine convolution (PCC) are proposed to refine feature representations near the boundary. Finally, the loss function is introduced to train the network in a supervised way. As a result, BMNet can achieve precise and detailed segmentation.

A. Overview

As shown in Fig. 1, the proposed BMNet mainly consists of two stages: the encoding stage where the feature representations of input images are extracted and the refinement stage where features are refined by the SAM and BGM, and then decoded as the predicted results.

First, in the encoding stage, the high-resolution network (HRNet) [43] is adopted as the backbone for semantic segmentation. As shown in encoding stage of Fig. 1, the HRNet maintains a high-resolution feature in the process of generating high-to-low-resolution features, which guarantees its excellent performance in feature extraction. Given an image I as the input, the encoder generates multiscale features via the backbone. Different from the original HRNet, the original classifier is abandoned and replaced with a 1×1 convolution after concatenating features from parallel branches, which is aimed to preliminarily fuse the multiscale features and match the input size of the refinement phase. The output F^{en} of the encoding phase is given by

$$F^{\text{en}} = f^{\text{en}}(I) \tag{1}$$

where $f^{\text{en}}(\cdot)$ denotes the operations of the encoder.

Second, in the refinement stage, we apply a structure of multiple parallel branches to enrich the scale diversity of features. Each branch contains two proposed modules: a BGM

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and an SAM, which are in parallel. The two modules in the ith branch have their own scale factors, which are named s_i^{sa}, d_i^{bg}, respectively. The s_i^{sa} controls the downsampling rate of the convolution layer before the SAM. Similarly, d_i^{bg} controls the dilation rate of the convolution layer before the BGM. By setting different scale factors for each branch, the proposed modules can obtain features with diverse receptive fields as inputs, which is helpful for the proposed modules to reason about features with various scales and enhance the scale diversity of features. The output of every branch is obtained by concatenating the outputs from the two proposed modules. Therefore, given the scale factor pair $(s_i^{\text{sa}}, d_i^{\text{bg}})$ of the ith branch, the output F_i^{re} of the branch is represented as

$$\begin{cases} F_i^{\text{en}} = \text{conv}\left(F^{\text{en}}\right) \\ F_i^{\text{sa}} = f_i^{\text{sa}}\left(F_i^{\text{en}}, s_i^{\text{sa}}\right) \\ F_i^{\text{bg}} = f_i^{\text{bg}}\left(F_i^{\text{en}}, d_i^{\text{bg}}\right) \\ F_i^{\text{re}} = \text{conv}\left(\text{concat}\left(F_i^{\text{sa}}, F_i^{\text{bg}}\right)\right) \end{cases} \tag{2}$$

where conv$(\cdot)$ is the 1×1 convolution operation, concat is the concatenation operation, and $f_i^{\text{sa}}(\cdot)$, $f_i^{\text{bg}}(\cdot)$ denote the BGM and SAM in the ith branch, respectively. In this stage, there exist four parallel branches with different scale factors pairs to enhance the scale diversity of features, so the refined feature output by the refinement stage is given by

$$F^c = \text{conv}\left(\text{concate}\left(\left\{F_i^{\text{re}}\right\}_{i=0}^{3}\right)\right) \tag{3}$$

where the feature set $\{F_i^{\text{re}}\}_{i=0}^{3}$ is the output from the above four parallel branches.

Finally, the feature F^c from the refinement stage is directly fed into the classifier cls$(\cdot)$ of the segmentation to generate the final prediction of segmentation P^{seg}. Therefore, the final prediction can be described as

$$P^{\text{seg}} = \text{cls}\left(F^c\right). \tag{4}$$

In the BMNet, multiscale features are extracted in parallel branches, which is necessary for a comprehensive understanding of the input image in a multiscale manner.

B. SAM

Objects often take on a variety of forms in images due to the scale variation and the noise from shadows, occlusions, backgrounds, and so on. The issue of scale variation is alleviated to some extent through multiscale parallel structures. However, the above noise may be mixed with features both in space and channel through convolution layers and leads to a decrease in the discriminability of features. Therefore, it is necessary to enhance the discriminability of multiscale features. For this purpose, we propose an efficient module, named SAM. The SAM constructs multiscale contextual dependencies to remove the noise in features. The SAM can reduce the intraclass distance and increase the interclass distance while maintaining the scale diversity of features, which makes features more discriminative. We illustrate the SAM in Fig. 2.

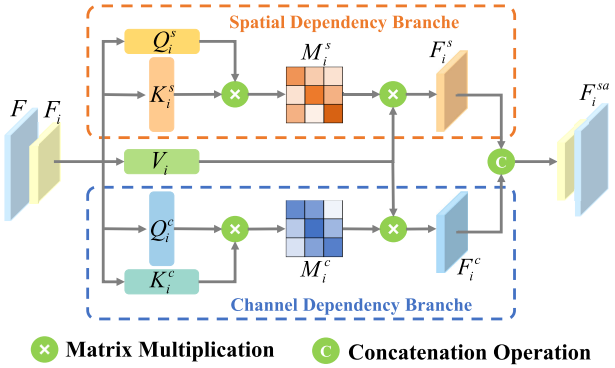


Fig. 2. Details of the SAM.

First, given the input feature $F \in \mathbb{R}^{C\times H\times W}$, the scaled feature $F_i \in \mathbb{R}^{C_i\times H_i\times W_i}$ is generated by downsampling with the scale factor as s_i, where $C, H, and W$ represent the channel, height, and width of features, respectively. Downsampling is implemented via convolution layers to enrich the scale diversity of features in the branch.

Second, to construct contextual dependencies in the spatial and channel dimensions, the scaled feature F_i is fed into four 1×1 convolution layers to obtain query maps and key maps both in space and channel $\{Q_i^s, K_i^s, Q_i^c, K_i^c\}$, respectively, and we can calculate the spatial contextual dependency $M_i^s \in \mathbb{R}^{H_iW_i\times H_iW_i}$ and the channel contextual dependency $M_i^c \in \mathbb{R}^{C_i\times C_i}$ as

$$\begin{cases} M_i^s = \text{softmax}\left(f_r\left(Q_i^s\right)^T \cdot f_r\left(K_i^s\right)\right) \\ M_i^c = \text{softmax}\left(f_r\left(Q_i^c\right) \cdot f_r\left(K_i^c\right)^T\right) \end{cases} \tag{5}$$

where $(\cdot)^T$ is the transpose operation, $f_r : \mathbb{R}^{C\times H\times W} \to \mathbb{R}^{C\times HW}$ is the reshape operation, and $\text{softmax}(M) = (e^{M_{mn}}/(\sum_{n=1}^{N} e^{M_{mn}}))$. The element $(M_i^s)_{mn}$ of M_i^s measures the correlation between the mth and nth pixels in F_i. Obviously, due to downsampling, a pixel in the scaled feature F_i represents a local region of the input feature F, which means the element $(M_i^s)_{mn}$ reasons about the relationship of local regions in the input feature F at the scale s_i, and a higher value of $(M_i^s)_{mn}$ indicates a stronger dependence between the two local regions. Similar to the spatial contextual dependency M_i^s, the channel contextual dependency M_i^c reasons about the relationship between channels at the scale i. The above relationships can effectively reduce the influence from the object scale variations, which makes features more discriminative.

Finally, to obtain long-range contextual features at the scale s_i, the value map V_i is obtained by applying a 1×1 convolution to the scaled feature $\hat{F}_i$. The value map V_i is shared by the spatial and channel contextual dependencies, which can improve the compatibility of features and makes it easier to be fused. With the help of the contextual dependencies $M_i^s and M_i^c$, the scaled contextual feature F_i^{sa} is generated as

$$\begin{cases} F_i^s = f_R\left(f_r(V_i) \cdot \left(M_i^s\right)^T\right) \\ F_i^c = f_R\left(M_i^c \cdot f_r(V_i)\right) \\ F_i^{\text{sa}} = \text{conv}\left(\text{up}\left(\text{concat}\left(F_i^s, F_i^c\right)\right)\right) \end{cases} \tag{6}$$

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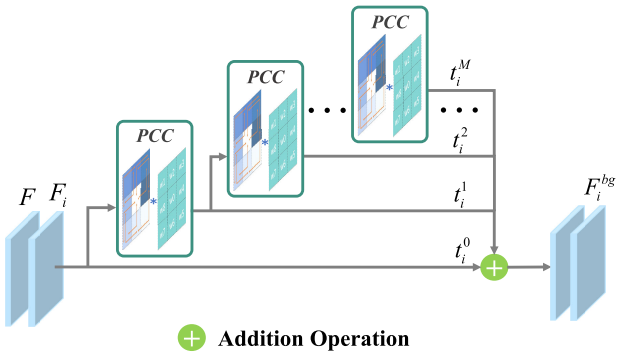


Fig. 3. Details of the BGM.

where conv(·) denotes a 1×1 convolution layer, up(·) : $\mathbb{R}^{C_i \times H_i \times W_i} \rightarrow \mathbb{R}^{C_i \times H \times W}$ is the upsampling operation, concat(·) is the concatenation operation, and $f_R : \mathbb{R}^{C \times HW} \rightarrow \mathbb{R}^{C \times H \times W}$ is the reshape operation. Fusing the features F_i^s and F_i^c generated by the contextual dependencies M_i^s and M_i^c, the feature F_i^{sa} has both the spatial and channel contextual dependencies, which makes itself more discriminative at the scale s_i.

The SAM aggregates long-range contextual information in a multiscale way, which greatly enhances the discriminability of features. With the SAM, our network can classify objects at different scales in an efficient and robust manner.

C. BGM

The SAM provides the multiscale dependencies to enhance the discriminability of features, but convolution layers are unable to explicitly extract boundary information, which results in an irreversible loss of boundary information and a poor distinction of boundaries of objects. Therefore, it is not reasonable to fuse multiscale features without boundary information for segmentation. Besides, multiscale features usually have different complexity of boundaries, and it is also a problem to be solved how to introduce boundary information into multiscale features to guide the recovery features near the boundary.

To deal with these issues, we propose an efficient module, called the BGM, which is illustrated in Fig. 3. The key component of the BGM is a novel operator called the PCC operator, which is shown in Fig. 4. The PCC takes advantage of the manifold information in features to explicitly encode the BG map. The BGM uses a BG map extracted by the PCC to refine features in a multilevel manner. Therefore, we first introduce the proposed PCC operator and then build the BGM with the PCC operator in the following.

1) Pixel Cosine Convolution: From a geometric point of view, the feature $F \in \mathbb{R}^{C \times H \times W}$ can be considered as the function $\mathcal{F}(h, w) = F(:, h, w)$, which constructs a discrete surface $\mathcal{S} \in \mathbb{R}^{C+2}$ by the graph $(h, w, F(:, h, w))$. Thus, let $p = (h_p, w_p, F(:, h_p, w_p))$ be a pixel on $\mathcal{S}$, and the angle θ_p is defined as the angle formed by the w-O-h coordinate plane and the tangent plane of $\mathcal{S}$ at p. Intuitively, the larger the angle θ_p, the greater the probability that the pixel p is at the boundary, which means θ_p can measure the boundary confidence of p. Therefore, the BG map G is given by

$$G(p) = \frac{1}{\cos\theta_p} = \frac{S(d\mathcal{S}_p)}{S(d\overline{\mathcal{S}}_p)} \tag{7}$$

where $S(d\mathcal{S}_p)$ is the area of the microsurface $d\mathcal{S}_p$ at p, and $S(d\overline{\mathcal{S}}_p)$ is its projected area on the w-O-h coordinate plane.

Since the surface $\mathcal{S}$ is discrete and is composed of countless individual pixels, the area of the microsurfaces $S(d\mathcal{S}_p)$ cannot be calculated directly. To solve this problem, the concept of manifolds, a topological space that can be locally Euclidean spaced, is borrowed from the discrete differential geometry (DDG). In DDG, the surface $\mathcal{S}$ is a 2-D manifold in a high-dimensional space, which is composed of countless microplanes, and each microplane is in a local Euclidean space. Therefore, the microsurface $d\mathcal{S}_p$ is approximated by the microplane, and its area can be approximated by the area of microplanes around the pixel p in a way of interpolation. To obtain the microplane, the pixel pair P_i, which forms the ith microplane around the pixel p, is defined as $P_i = (p_i, p_{ih}, p_{iw})$. And $S(d\mathcal{S}_p)$, $S(d\overline{\mathcal{S}}_p)$ can be expressed as follows:

$$\begin{cases} S(d\mathcal{S}_p) = \sum\limits_{P_i \in \mathcal{P}_K} \omega_i \cdot S(\mathcal{S}_{P_i}) \\ S(d\overline{\mathcal{S}}_p) = \sum\limits_{P_i \in \mathcal{P}_K} \omega_i \cdot S(\overline{\mathcal{S}}_{P_i}) \end{cases} \tag{8}$$

where ω_i is the learnable interpolation weight, and K is the neighborhood diameter. $S(\mathcal{S}_{P_i})$ is the area of the ith microplane around the pixel p. $S(d\overline{\mathcal{S}}_p)$ is its projected area on the w-O-h coordinate plane. $\mathcal{P}_K = \{P_i = (p_i, p_{ih}, p_{iw}) | i \in [0, 1, \ldots, N], N \leq K^2 - 1\}$ is a set of pixel pairs belonging to the K-neighborhood of the pixel p, which means the first element p_i of the P_i is selected from the current local patch of size $K \times K$. To form the microplane, the pixel pair $P_i = (p_i, p_{ih}, p_{iw})$ satisfies the following:

$$\begin{cases} \nabla h_i = |h_{p_i} - h_{p_{ih}}| = r \\ \nabla w_i = |w_{p_i} - w_{p_{iw}}| = r \\ |w_{p_i} - w_{p_{ih}}| = 0 \\ |h_{p_i} - h_{p_{iw}}| = 0 \end{cases} \tag{9}$$

where p_i, p_{ih}, p_{iw} are the pixels of $\mathcal{S}$, and r is the stride factor, which controls the size of the microplanes $\mathcal{S}_{P_i}$ and satisfies $r \leq K$. Obviously, the projected area $S(\overline{\mathcal{S}}_{P_i})$ can be easily obtained by

$$S(\overline{\mathcal{S}}_{P_i}) = \frac{\nabla h_i \nabla w_i}{2} = \frac{r^2}{2}. \tag{10}$$

And according to the plane projection theorem, we can obtain

$$\begin{aligned} S(\mathcal{S}_{P_i}) &= S(\overline{\mathcal{S}}_{P_i}) \sqrt{\left(\frac{\nabla F_{P_i}^h}{\nabla h_i}\right)^2 + \left(\frac{\nabla F_{P_i}^w}{\nabla w_i}\right)^2 + 1} \\ &= \frac{r^2}{2} \sqrt{\left(\frac{\nabla F_{P_i}^h}{r}\right)^2 + \left(\frac{\nabla F_{P_i}^w}{r}\right)^2 + 1} \end{aligned} \tag{11}$$

where $\nabla F_{P_i}^h = F(:, h_{p_{ih}}, w_{p_{ih}}) - F(:, h_{p_i}, w_{p_i})$ is the pixel difference of the pixel pair P_i in height, and $\nabla F_{P_i}^w =$

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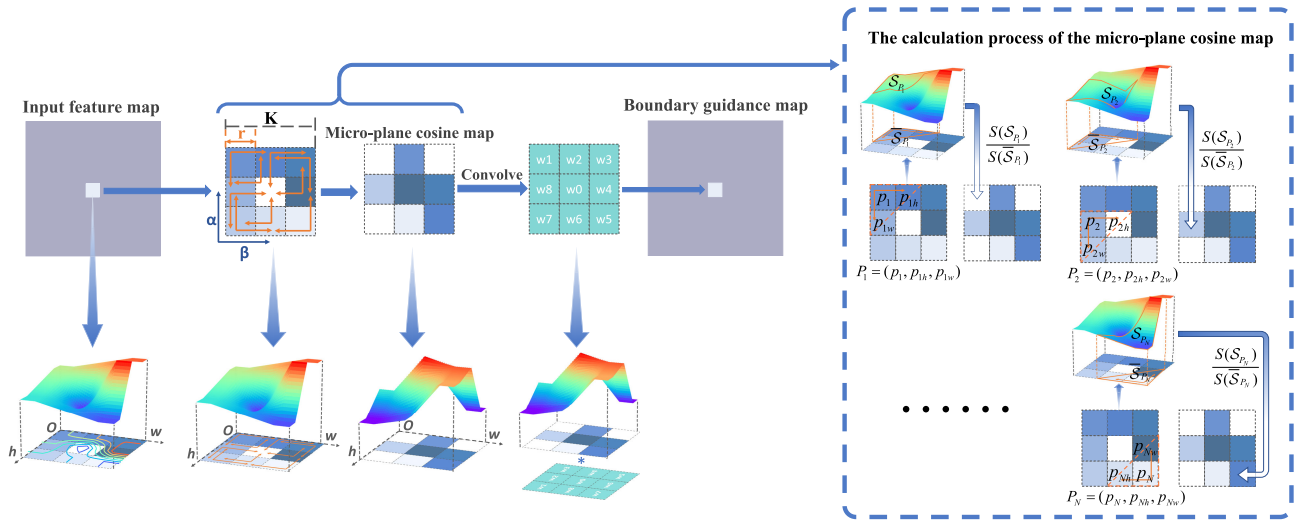


Fig. 4. Diagrams of PCC. Pixels passed by a orange cross-arrow represent a pixel pair P_i which can form a microplane $\mathcal{S}_{P_i}$, and the cosine of the microplane is calculated and filled in the position where the orange cross-arrows cross to generate the microplane cosine map. Then, the microplane cosine map is convolved with convolution kernel and outputs the BG map.

$F(:, h_{p_{iw}}, w_{p_{iw}}) - F(:, h_{p_i}, w_{p_i})$ is the pixel difference of the pixel pair P_i in width. Combining (8), (10), and (11), the BG map G can be formulated as

$$G(p) = \sum_{P_i \in \mathcal{P}_K} \omega_i \frac{S(\mathcal{S}_{P_i})}{S(\overline{\mathcal{S}}_{P_i})} = \sum_{P_i \in \mathcal{P}_K} \frac{\omega_i}{cos(\theta_{P_i})} = \sum_{P_i \in \mathcal{P}_K} \omega_i \sqrt{\left(\frac{\nabla F^h_{P_i}}{r}\right)^2 + \left(\frac{\nabla F^w_{P_i}}{r}\right)^2 + 1} \tag{12}$$

where θ_{P_i} is the angle formed by the w-O-h coordinate plane and the microplane $S(\mathcal{S}_{P_i})$.

Equation (12) shows the calculation process of the proposed PCC operator, and the operator explicitly encodes the boundary information from the perspective of the manifold. Considering the scale difference both in width and height, $\mathcal{S}$ should be scaled by (α/r) in height and (β/r) in width. Finally, the PCC operator is rewritten as

$$\mathrm{PCC}(p, r, K) = \sum_{P_i \in \mathcal{P}_K} \omega_i \sqrt{\left(\frac{\nabla F^h_{P_i}}{\alpha}\right)^2 + \left(\frac{\nabla F^w_{P_i}}{\beta}\right)^2 + \delta} \tag{13}$$

where δ is the bias term, and α, β, δ are all trainable parameters to ensure scale adaptiveness. When $K = 1, \alpha = \beta = 1, \delta = 0$, the PCC degrades to the following form:

$$\mathrm{PCC}(p, r = 1, K = 1) = \sqrt{\left(\nabla F^h_P\right)^2 + \left(\nabla F^w_P\right)^2} \tag{14}$$

which is exactly the formula of the gradient at the pixel p. Therefore, the gradients of the feature F is a special case of the PCC.

2) Constructing BGM With PCC: To adaptively introduce boundary information into multiscale features and refine features near the boundary, we build the BGM with the PCC operator. The feature refinement process of the BGM is considered as the time-varying function $\mathcal{G}(t, h, w)$, and the input feature after scaling $F_i \in \mathbb{R}^{C \times H \times W}$ is the initial state of the function $\mathcal{G}$, i.e., $\mathcal{G}(0, h, w) = F_i(:, h, w)$. According to the Taylor series, any function $g(t)$ can be approximated as the summation of its multiorder derivatives

$$g(t) = \sum_{n=0}^{+\infty} \frac{1}{n!} g^{(n)}(t_0)(t - t_0)^n. \tag{15}$$

Therefore, the refined feature F_i^{bg} is considered as the value of $\mathcal{G}$ at moment t_i, i.e., $F_i^{\mathrm{bg}}(:, h, w) = \mathcal{G}(t_i, h, w)$, the PCC operator is used to extract multiorder boundary information, and F_i^{bg} is given by

$$\begin{cases} F_i = d\mathrm{conv}(F) \\ F_i^{\mathrm{bg}} = \mathrm{conv}\left(\sum_{m=0}^{M} t_i^m \mathrm{PCC}^{(m)}(F_i)\right) \\ \mathrm{PCC}^{(0)}(F_i) = F_i \end{cases} \tag{16}$$

where F is the input feature. $d\mathrm{conv}(\cdot)$ denotes the dilated convolution, which makes the scaled features F_i have different scale resolutions. $\{t_i^m\}_{m=0}^M$ are trainable weight parameters. m denotes the number of the PCC operator, which determines the accuracy of the boundary information and satisfies $\mathrm{PCC}^{(m)}(F_i) = \mathrm{PCC}(\mathrm{PCC}^{(m-1)}(F_i))$. In this process, the multiscale features near the boundaries are refined adaptively, which is beneficial to improve the performance in semantic segmentation.

D. Loss Function

Similar to prevalent methods for segmentation tasks, the proposed network BMNet is trained in an end-to-end manner. Given an image, the network predicts a segmentation mask P^b. For this task, it is a common choice to apply the cross-entropy loss L on predictions

$$L = \frac{1}{B}\sum_{b=1}^{B} L_{\mathrm{ce}}\left(P^b, G^b\right) = \frac{1}{B}\frac{1}{N}\sum_{b=1}^{B}\sum_{n=1}^{N} \left|G_n^b \log\left(\mathrm{softmax}\left(P_n^b\right)\right)\right|_1 \tag{17}$$

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where $L_{ce}(\cdot,\cdot)$ is the cross-entropy loss, P_n^b, G_n^b is the predicted mask and its label of the nth pixel in the bth image, $|\cdot|_1$ is the $L1$-norm, B is the number of training images, and $N = H \times W$ is the number of pixels in an image.

IV. Experiments

In this section, two remote sensing datasets are used to validate the performance of the proposed network. First, the datasets and implementation details about experiments are introduced. Second, the effectiveness of the proposed method is studied by ablation experiments on two datasets. Finally, we provide comparisons with other several prevalent methods and the benchmark on the datasets.

A. Data Description

1) Potsdam Dataset: This dataset is a public remote sensing dataset provided by the International Journal of Photogrammetry and Remote Sensing (ISPRS). The dataset contains a total of 38 tile aerial images with a ground sample distance (GSD) of 5 cm, and each image contains the true orthophoto (TOP) tile and its corresponding normalized digital surface model (nDSM) tiles with a uniform size of 6000 × 6000 pixels. The TOP tiles contain four channels of *near-infrared*, *red*, *green*, and *blue*. All the images are labeled with common six land-cover categories, including *impervious surfaces*, *building*, *low vegetation*, *tree*, *car*, and *background*. In our experiments, 24 tiles of the dataset are used for training, and the remaining 14 tiles are used for testing, which is the same arrangement as Commission III of the ISPRS.

2) Vaihingen Dataset: Similar to the Potsdam dataset, this dataset is also a widely used remote sensing dataset provided by ISPRS. The dataset contains a total of 33 tile aerial images with a GSD of 9 cm, and each image contains the TOP tile and its corresponding nDSM tiles. Every tile in the dataset has a different size with an average of 2494 × 2064 pixels. The TOP tiles contains only three channels of *near-infrared*, *red*, and *green*. All the images are also labeled with the same six land-cover categories as that of the Potsdam dataset. Following the arrangement with Commission III of the ISPRS, 16 tiles of the dataset are used for training and the remaining 17 tiles for testing.

B. Implementation Details

1) Data Preprocessing: To avoid overfitting problems due to unbalanced distributions of samples, it is necessary to perform data augmentation on samples. Therefore, each image is first cropped to patches with a size of 512 × 512 pixels. After that, we randomly resize the patches between 0.5 and 2 times and then randomly flip them with a probability of 0.5 in the training phase. During the testing phase, the cropping and flipping operations are applied to each test image. The final prediction results are calculated by averaging the results of the original and flipped images.

TABLE I
Effectiveness of Spatial Scale Combinations on Two Datasets

Dateset	SA-Module		BG-Module			m-F_1(%)	OA(%)
	S_1	S_2	D_1	D_2	D_3		
Potsdam	✓		✓			92.8	91.3
	✓			✓		**93.2**	**91.8**
	✓				✓	92.9	91.5
		✓	✓			92.7	91.3
		✓		✓		93.0	91.6
		✓			✓	92.8	91.5
Vaihingen	✓		✓			89.7	90.7
	✓			✓		**90.3**	**91.0**
	✓				✓	89.9	90.8
		✓	✓			89.6	90.6
		✓		✓		90.0	90.8
		✓			✓	89.7	90.6

2) Experimental Setting: In our experiments, the proposed BMNet is implemented with PyTorch. The HRNet pretrained on the ImageNet is applied as our backbone and baseline of the proposed network during the entire experiments for all the datasets. In the training phase, we adopt SGD as our optimizer with a momentum of 0.9 and a weight decay of 0.0005. The cosine learning decay policy is applied to control the change in the learning rate, which is described as follows:

$$\text{cur_}lr = \frac{\text{init_}lr}{2}\left(1 + \pi \cdot \cos\left(\frac{\text{cur_iter}}{\text{max_iter}}\right)\right) \tag{18}$$

where init_lr is the initial learning rate and is set to 0.003 for the Potsdam dataset and 0.004 for the Vaihingen dataset, cur_iter is the index of the current iteration, max_iter is the maximum index of iterations, and cur_lr is the learning rate of the current iteration. We set 100 epochs for the Potsdam dataset and 200 for the Vaihingen dataset. The batch size is set to 8 per GPU for all the datasets. All the experiments are run on two NVIDIA GeForce RTX 2080Ti GPUs.

3) Evaluation Metrics: To quantify and compare the experimental performance more comprehensively, we choose two widely used metrics of semantic segmentation tasks: overall accuracy (OA) and F_1 score, which evaluate the accuracy of the prediction results in different ways. First, OA reports the proportion of samples with correct predictions to the total samples

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}. \tag{19}$$

Second, the F_1 score reports the comprehensive evaluation of the accuracy $P = \text{TP}/(\text{TP} + \text{FP})$ and the recall $R = \text{TP}/(\text{TP} + \text{FN})$

$$F1 = \frac{2PR}{P + R} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}} \tag{20}$$

where TP, FP, TN, and FN are the true positive, the false positive, the true negative, and the false negative, respectively. The mean F_1 (m-F_1) score is calculated by averaging the F_1 scores over all categories except the background category to serve as a comprehensive evaluation.

4) Hyperparameter Setting: As shown in 2, the scale factor pair $(s_i^{\text{sa}}, d_i^{\text{bg}})$ controls the spatial scale combinations of the SAM and BGM in the ith branch. Specifically, s_i^{sa} controls the

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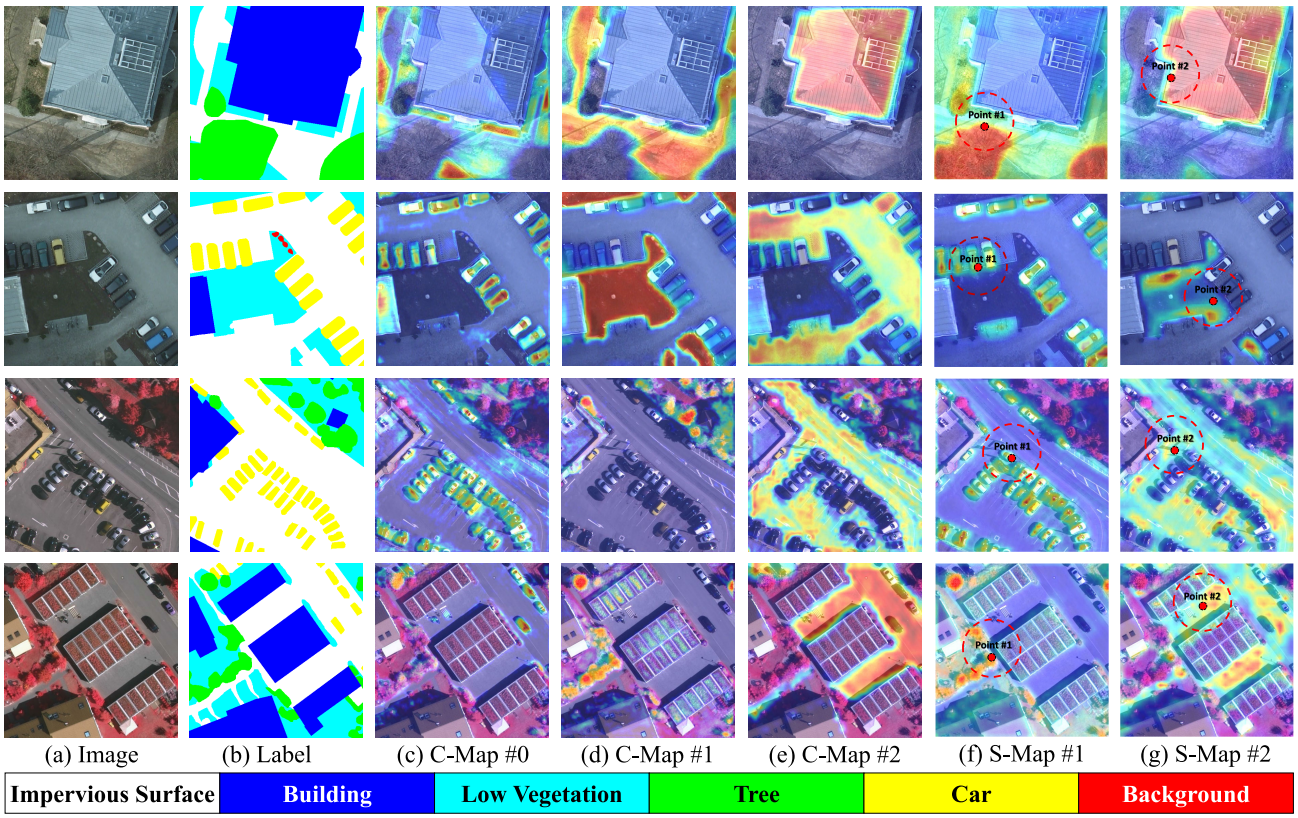


Fig. 5. Visualization of features from the SAM on the Potsdam and Vaihingen test sets. For each row, we show an input image, ground truth in (a) and (b), three channel maps (C-maps corresponding to F_i^c in the SAM) from the outputs of channel contextual dependencies at different three scales (scale #0, scale #1, and scale #2) in (c)–(e), and two spatial contextual dependencies (S-maps corresponding to M_i^s in the SAM) at different two scales (scale#1 and scale#2) in (f) and (g), which represents the relationship between the marked red pixel circled by red dashed lines and other pixels in the image.

downsampling rate of the SAM, and d_i^{bg} controls the dilation rate of the BGM. The hyperparameter $(s_i^{\text{sa}}, d_i^{\text{bg}})$ is not robust to the segmentation task, and different scale combinations may have an influence on the segmentation performance of the proposed BMNet. Therefore, it is necessary for the segmentation task to set the optimal scale combination. Since the proposed BMNet has four parallel branches in the refinement stage, there are four scale factor pairs to be set. To simplify the expression, the above hyperparameters are divided into two groups, $S = (s_1^{\text{sa}}, s_2^{\text{sa}}, s_3^{\text{sa}}, s_4^{\text{sa}})$ and $D = (d_1^{\text{bg}}, d_2^{\text{bg}}, d_3^{\text{bg}}, d_4^{\text{bg}})$.

Table I shows the effectiveness of spatial scale combinations on segmentation performance. The hyperparameter S is set to two options $S_1 = ((1/4), (1/8), (1/16), (1/16))$ and $S_2 = ((1/4), (1/8), (1/16), (1/32))$, where the downsampling rate is with respect to the size of the input image. The hyperparameter D is set to three options $D_1 = (1, 1, 1, 1)$, $D_2 = (1, 2, 2, 4)$, and $D_3 = (1, 2, 4, 8)$. It is obvious that OA and m-F_1 score have a great performance while $S = S_1$ and $D = D_2$ in Table I. Besides, when S remains the same, $D = D_2$ is the best choice for the segmentation performance, and the performance starts to decrease with the continued increase in dilation rate. When D remains the same, $S = S_1$ is the best choice. So the high resolution of features is beneficial for the segmentation performance, but, on the contrary, it will have a higher computational cost. In summary, the two hyperparameter groups $S and D$ are independent of each other to some extent, and the optimal option for segmentation performance is $S = S_1 and D = D_2$ in all the experiments.

C. Effectiveness Analysis of SAM

1) Visualization Analysis: To intuitively explore the role the SAM played in feature representations, we visualize some features of the SAM at different scales in Fig. 5. The brightness of the region represents how much attention the module pays to the region, and the brighter the region, the more interested the module is in the region. Thereby, we can note how the SAM refines feature maps of different scales.

First, we visualize the channel maps refined by channel contextual dependencies at different scales. As shown in Fig. 5(c), the channel maps from scale #0 (1/4 downsampling) pay more attention to small-scale cars. Fig. 5(d) shows that middle-scale trees and low vegetations are in focus by the channel maps from scale #1 (1/8 downsampling). Large-scale impervious surfaces and buildings obtained high responses in the channel maps from scale #2 (1/16 downsampling) in Fig. 5(e). The above visualizations demonstrate that channel contextual dependencies of the SAM can perceive objects from various scales, which is beneficial for the segmentation of multiscale objects. Next, we visualize the spatial contextual dependencies at different scales in the marked red pixel, which represents the relationship between the marked red pixel and other pixels in the image. Fig. 5(f) shows the spatial contextual dependency of small objects such as cars and trees at scale #1 (1/8 downsampling), and the spatial contextual dependency at scale #2 (1/16 downsampling) extracts the relationships of large objects such as buildings and impervious surfaces in Fig. 5(g). Therefore, spatial contextual dependencies of the

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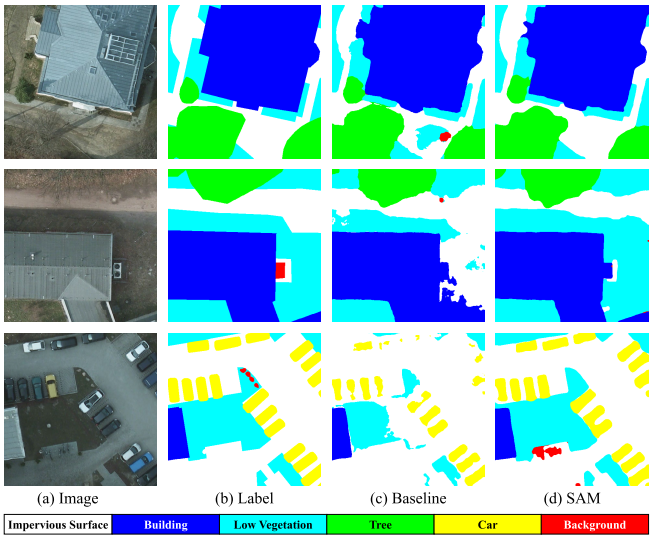


Fig. 6. Comparison of segmentation results with and without the SAM on the Potsdam test set. (a) Image. (b) Label. (c) Baseline. (d) SAM.

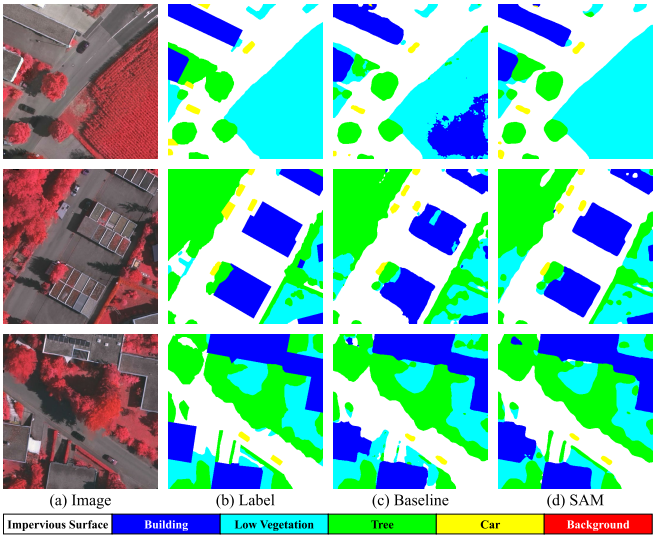


Fig. 7. Comparison of segmentation results with and without the SAM on the Vaihingen test set. (a) Image. (b) Label. (c) Baseline. (d) SAM.

SAM are effective for integrating the information of categories, which is beneficial for reducing the intraclass distance and increasing the interclass distance. In short, the SAM makes features more discriminate with multiscale contextual dependencies.

2) Ablation Analysis: To verify the effectiveness of the SAM, we present comparisons of segmentation performance obtained by the proposed modules on datasets, as shown in Table II. It is obvious that the SAM improves the m-F_1 score from 91.8% to 92.6% (0.8% ↑) on the Potsdam dataset and from 88.6% to 89.5% (0.9% ↑) on the Vaihingen dataset, and it improves the OA score from 90.4% to 91.2% (0.8% ↑) on the Potsdam dataset and from 90.0% to 90.6% (0.6% ↑) on the Vaihingen dataset. The improvement of segmentation performance on datasets shows that the SAM possesses the ability to enhance the discriminability of multiscale features and obtain a great result on remote sensing datasets.

Fig. 6 shows some of the segmentation results of applying the SAM on the Potsdam dataset. Multiscale variation in objects plays an important role in the category confusion in segmentation. With the help of the SAM, multiscale contextual dependencies are constructed, which makes features more discriminative. As shown in the first row of Fig. 6, the HRNet (baseline) misclassifies the internal region of the impervious surfaces as the low vegetation. While the SAM effectively reduces the intraclass distance and corrects the misclassifications of internal regions. As shown in the second and third rows of Fig. 6, adjacent regions of different categories are misclassified over a wide range, such as the impervious surfaces and the low vegetation, the building and the low vegetation, and the car and the impervious surfaces. After applying the SAM, the interclass distance is increased, which makes the prediction results of adjacent regions improved. Therefore, the SAM can adaptively refine the effective information from multiscale features and greatly improve the feature representation of objects.

However, misclassification of the background (red pixels) still occurs after applying the SAM in Fig. 6. There are two main reasons for the misclassification. First, the background has too large an intraclass variance to discriminate. As shown in the second row of Fig. 6, the region labeled as the background is clearly more similar to the buildings, and the above region is unsurprisingly misclassified as the buildings. Second, the misclassification of the background is also attributed to the coarse ground truth. As shown in the third row of Fig. 6, the region labeled as the impervious surface is misclassified as the background due to the bicyclists and pedestrians on the impervious surfaces, which are completely ignored.

We also test the SAM on the Vaihingen dataset and show some of the segmentation results in Fig. 7. Due to the variations in the appearance of objects, the category confusion appears more frequently in the segmentation results without the SAM. The intraclass noise leads to confusion in internal regions between the building and the low vegetation in the first and second rows of Fig. 7. And the interclass noise results in the confusion of adjacent regions between the impervious surfaces, the tree, and the low vegetation in the third row of Fig. 7. Comparing before and after the SAM, it is evident that the SAM can eliminate these noises and improve the performance in segmentation.

D. Effectiveness Analysis of BGM

1) Visualization Analysis: To understand how the BGM influences the segmentation results, we visualize the BG maps generated by the PCC and the feature maps output by the BGM in Fig. 8. To observe the visualizations more intuitively, we mix them with their own original images.

Different dilation rates of the BGM make itself have a different concern. When the BGM sets the dilation rate as dilation #0 (dilation rate = 1), the PCC effectively extracts the boundary information of small-scale cars in Fig. 8(c) and refines representations near boundaries in Fig. 8(f). As shown in Fig. 8(d), the boundary map of the middle-scale tree is captured via the BGM with dilation #1 (dilation rate = 2),

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TABLE II
ABLATION RESULTS (PER-CLASS F_1, m-F_1, AND OA) OF THE SAM AND BGM ON TWO DATASETS

Dateset	Method	SA-Module	BG-Module	F_1(%)					m-F_1(%)	OA(%)
				Imp. Surf.	Building	Low Veg.	Tree	Car		
Potsdam	Baseline			92.2	96.3	87.0	88.5	95.2	91.8	90.4
	BMNet-1	✓		93.4	97.3	87.6	88.4	96.3	92.6	91.2
	BMNet-2		✓	93.1	97.2	87.4	88.8	96.3	92.6	91.1
	BMNet-3	✓	✓	**93.7**	**97.5**	**88.5**	**89.5**	**96.9**	**93.2**	**91.8**
Vaihingen	Baseline			92.1	95.1	83.4	89.2	83.5	88.6	90.0
	BMNet-1	✓		92.3	95.7	84.3	89.5	85.7	89.5	90.6
	BMNet-2		✓	92.2	**95.8**	83.8	89.4	86.5	89.5	90.4
	BMNet-3	✓	✓	**92.5**	**95.8**	**85.3**	**90.0**	**88.0**	**90.3**	**91.0**

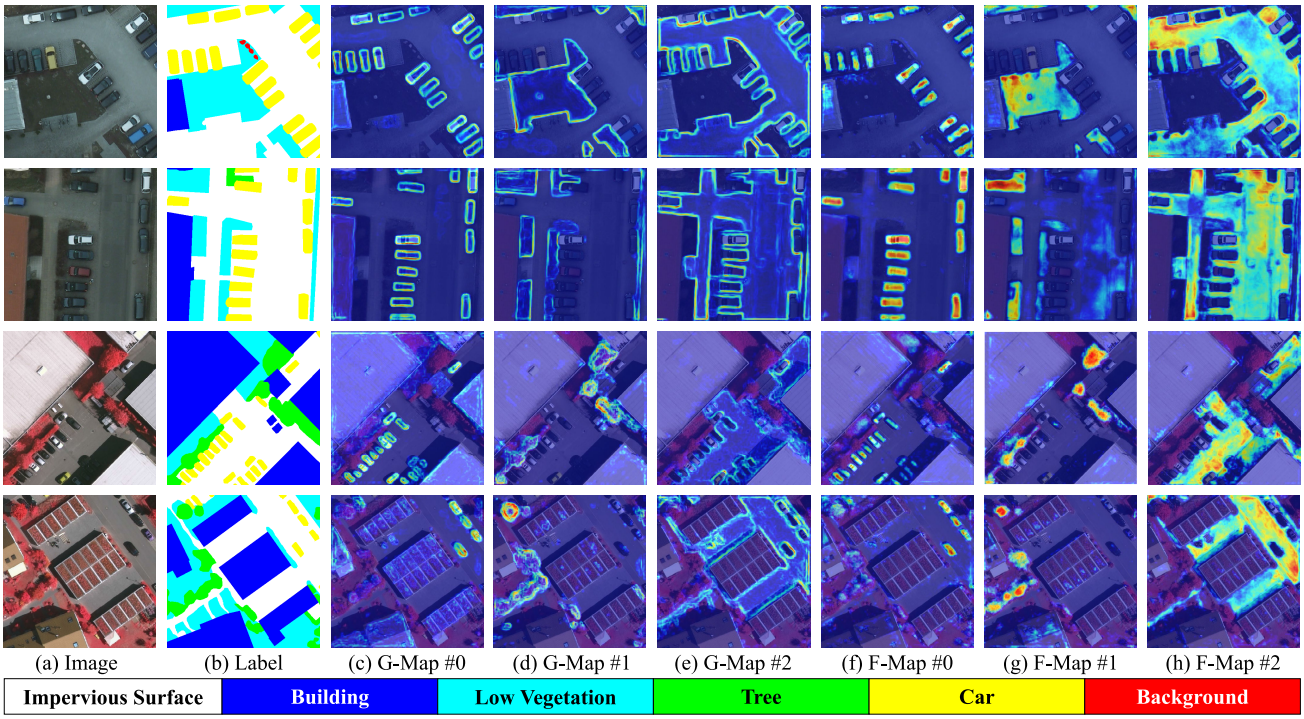


Fig. 8. Visualization of features from the BGM on the Potsdam and Vaihingen test sets. For each row, we show an input image, ground truth in (a) and (b), three BG maps from the output of the PCC with three different dilation rates (dilation #0, dilation #1, and dilation #2) in (c)–(e), and three feature maps from the output of the BGM at three different dilation rates (dilation #0, dilation #1, and dilation #2) in (f)–(h).

and the boundary of middle-scale tree becomes more accurate in Fig. 8(g). By setting the BGM to dilation #2 (dilation rate = 4), Fig. 8(e) shows concerns about the boundary map of the large-scale impervious surfaces, and the refined feature of the impervious surfaces is obtained in Fig. 8(h). Therefore, it can be qualitatively proven that the BGM can focus on the boundary of objects with the help of the PCC, and then guide the refinement of multiscale features at the boundary.

2) Ablation Analysis: As shown in Table II, it is verified that the BGM is beneficial to the improvement of performance in segmentation. Adding only the BGM, the m-F_1 score is raised from 91.8% to 92.6% (0.8% ↑) on the Potsdam dataset and from 88.6% to 89.5% (0.9% ↑) on the Vaihingen dataset, and the OA score is raised from 90.4% to 91.1% (0.7% ↑) on the Potsdam dataset and from 90.0% to 90.4% (0.4% ↑) on the Vaihingen dataset. Moreover, adding the BGM after the SAM, the m-F_1 score is raised by 0.6% to 93.2% on the Potsdam dataset and by 0.8% to 90.3% on the Vaihingen dataset, and the OA is raised by 0.6% to 91.8% on the Potsdam dataset and by 0.4% to 91.0% on the Vaihingen dataset.

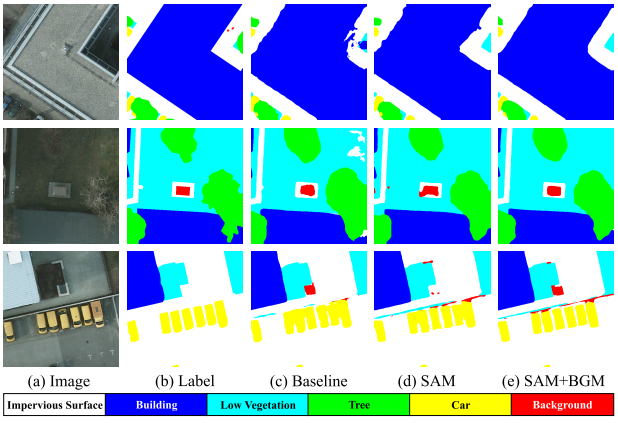


Fig. 9. Comparison of segmentation results with and without the BGM on the Potsdam test set. (a) Image. (b) Label. (c) Baseline. (d) SAM. (e) SAM + BGM.

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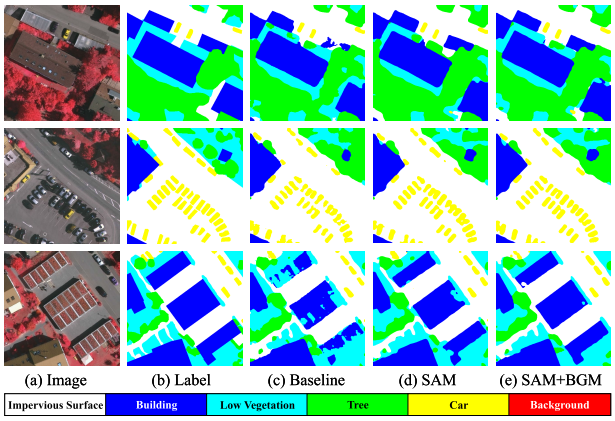


Fig. 10. Comparison of segmentation results with and without the BGM on the Vaihingen test set. (a) Image. (b) Label. (c) Baseline. (d) SAM. (e) SAM + BGM.

The above improvement in the performance in segmentation quantitatively verifies that the BGM refines the detail of objects at the boundaries.

To visually confirm the refinement in the segmentation results of the BGM, we compare the segmentation results with and without the BGM on the Potsdam dataset in Fig. 9. For HRNet, large-scale objects such as buildings are predicted with blurred boundaries, and small-scale objects such as cars have no sharp contours and cannot even be recognized in Fig. 9(c). Fig. 9(d) shows the segmentation results after applying the SAM. We can see that the SAM is helpless in refining the boundaries of multiscale objects and cannot solve the issue of locating boundaries. However, with the introduction of the BGM, the BGM supplies smoother boundaries for large-scale objects such as buildings, and sharper contours for small-scale objects such as cars in Fig. 9(e). The BGM can capture boundary information and guide networks to refine the representation at the boundary, which is the key to the improvement of performance in segmentations.

Fig. 10 shows the effect of the BGM on the Vaihingen dataset. It is obvious that the SAM generates better segmentation results than the HRNet. Long-range contextual dependencies from the SAM make multiscale features more discriminative. However, as shown in Fig. 10(d), ignorance of the geometry of objects makes it difficult to locate the boundaries of objects accurately. With the ability to extract boundary information from the BGM, networks eliminate the boundary noise, and the noise regions of objects such as buildings and cars are corrected in Fig. 10(e). After applying the SAM and the BGM, the BMNet can take into account the local representations of boundaries and the global representations of categories, which makes the segmentation results better.

E. Comparison With State-of-the-Art Methods

The segmentation performance of BMNet is compared with popular methods on the test set of datasets. Among comparison methods, one part of the methods is implemented with the ResNet101 as the backbone network, including FCN, UNet, SegNet, RefineNet, pyramid scene parsing network (PSPNet), CCEM, and DeepLabV3+, and the other part uses HRNet as the backbone network, including HRNet (baseline), HRNet-object-contextual representation (OCR), and our BMNet. To compare the above methods, these methods are set with their own hyperparameter settings, and the size of the test images is the same as that of the image images.

TABLE III
COMPARISON RESULTS (PER-CLASS F_1, m-F_1, AND OA) OF THE BMNET ON THE POTSDAM TEST SET

Method	F_1(%)					m-F_1(%)	OA(%)
	Imp. Surf.	Building	Low Veg.	Tree	Car		
FCN [36]	87.9	94.9	81.5	83.8	85.6	86.7	87.1
UNet [37]	88.6	93.3	83.3	85.8	88.5	87.9	86.6
SegNet [65]	89.0	93.7	86.3	84.9	89.2	88.6	87.9
RefineNet [66]	88.1	93.1	85.6	86.3	90.3	88.7	88.3
PSPNet [41]	91.6	96.3	86.4	86.8	91.4	90.5	89.5
CCEM [67]	89.7	94.5	86.9	85.6	92.0	89.8	89.9
DeepLabV3+ [40]	92.1	96.8	85.0	86.1	93.6	90.7	89.7
HRNet-OCR [68]	93.1	97.0	87.1	88.4	95.8	92.3	90.9
HRNet [43]	92.2	96.3	87.0	88.5	95.2	91.8	90.4
BMNet	**93.7**	**97.5**	**88.5**	**89.5**	**96.9**	**93.2**	**91.8**

1) Comparison on Potsdam Dataset: Table III shows the segmentation performance of various comparison methods on the Potsdam dataset. From Table III, we can observe that FCN is good at predicting large-scale buildings, but not small-scale objects such as trees due to the ignorance of details. By reusing high-level and low-level features to capture details, UNet improves the prediction of small-scale cars compared with FCN. SegNet improves nonlinear upsampling with max-pooling indexing, which is the key to obtaining better segmentation results than UNet. With the help of a chained residual pooling module, RefineNet can capture context more effectively and outperforms SegNet in cars. PSPNet is superior to RefineNet in impervious surfaces, thanks to the fact that its PPMs can obtain a global contextual prior. Through improving pooling operations with dilation convolutions in PPMs, DeepLabV3+ maintains the resolution of features and is better than PSPNet in segmentation performance. Compared with the methods mentioned above, HRNet (baseline) maintains high-resolution features throughout the encoding and decoding process, which means fewer details are lost. Therefore, there is no doubt that this method is superior to the methods mentioned above. Different from HRNet, HRNet-OCR feeds the output of HRNet into the OCR module to construct long-range contextual dependencies, so it can outperform HRNet.

For the proposed BMNet, it is obvious that it outperforms the above comparison methods, both in OA and the mean F_1 (m-F_1) score. Compared with HRNet (baseline), BMNet obtains gains of 1.4% both in the m-F_1 score and OA score. The F_1 scores of impervious surfaces, low vegetation, and cars all have a great improvement. The clues to the above improvement in segmentation performance can be found in Fig. 11, which shows examples of qualitative results of the Potsdam dataset. As shown in Fig. 11(c)–(g), there exist obvious misclassifications of impervious surfaces and low vegetation. Low vegetation has high intraclass variation and its sparseness on impervious surfaces will greatly affect the representation of both, particularly color and texture, which is

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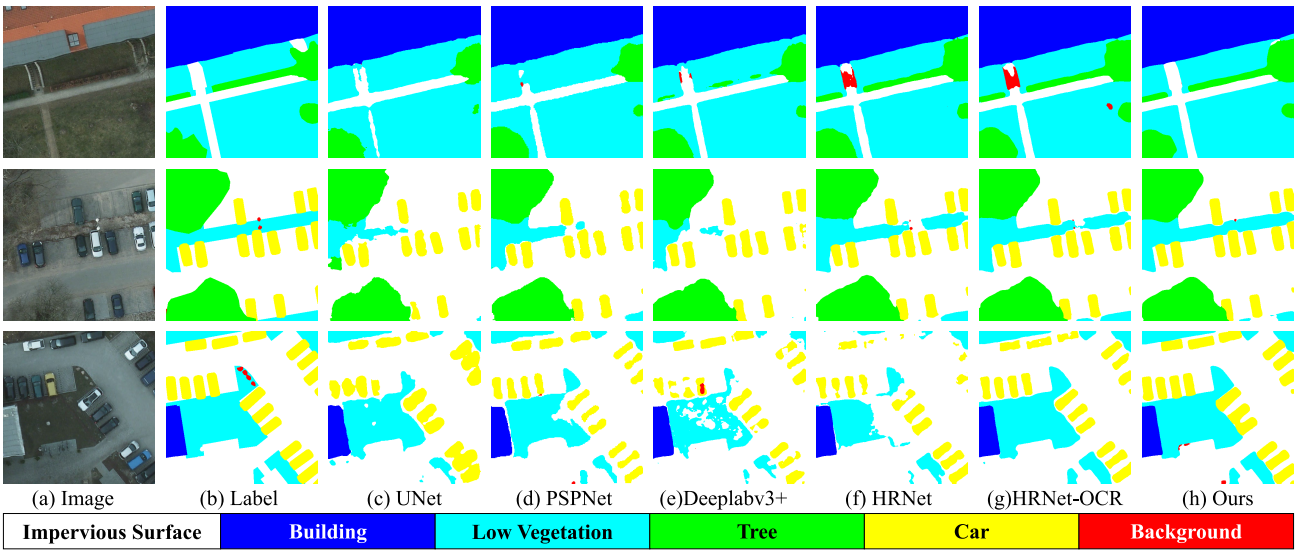


Fig. 11. Comparison of segmentation results between our method and prevalent methods on the Potsdam test set. (a) Image. (b) Label. (c) UNet. (d) PSPNet. (e) Deeplabv3+. (f) HRNet. (g) HRNet-OCR. (h) Ours.

TABLE IV
COMPARISON RESULTS (PER-CLASS F_1, m-F_1, AND OA) OF THE BMNET ON THE VAIHINGEN TEST SET

Method	F_1(%)					m-F_1(%)	OA(%)
	Imp. Surf.	Building	Low Veg.	Tree	Car		
FCN [36]	88.7	90.8	76.3	86.7	74.2	83.3	86.5
UNet [37]	90.5	93.3	79.6	87.5	76.4	85.5	89.2
SegNet [65]	90.2	93.7	78.5	85.8	83.9	86.4	88.5
RefineNet [66]	91.1	94.1	79.8	87.2	82.3	86.9	88.9
PSPNet [41]	90.6	94.3	79.0	87.0	70.7	84.3	89.1
CCEM [67]	91.5	93.8	79.4	87.3	83.5	87.1	88.6
DeepLabV3+ [40]	91.4	94.7	79.6	87.6	85.8	87.8	89.9
HRNet-OCR [68]	92.4	95.6	83.8	89.3	83.7	89.0	90.4
HRNet [43]	92.1	95.1	83.4	89.2	83.5	88.6	90.0
BMNet	**92.5**	**95.8**	**85.3**	**90.0**	**88.0**	**90.3**	**91.0**

a key factor in misclassification. Therefore, it is difficult to classify the above two categories correctly. The BMNet constructs spatial and channel contextual dependencies to reduce the intraclass variation and increase the interclass variation. As a result, impervious surfaces are well distinguished from the low vegetation in Fig. 11(h). As for cars, their small scale and dense arrangement cause the difficulty in classification, particularly in the boundaries, which can be seen in the last two rows of Fig. 11. However, the BMNet has the ability to capture boundary information, which can guide networks to locate the boundaries of objects. Thereby, as shown in Fig. 11(h), BMNet effectively eliminates the noise in the boundaries and does well in segmenting the details of objects.

2) Comparison on Vaihingen Dataset: To further demonstrate the superiority of the proposed BMNet, we also implemented comparison experiments with BMNet and some prevalent methods on the Vaihingen dataset. The segmentation results are summarized in Table IV. Unsurprisingly, the proposed BMNet is still superior to other comparison methods, both in the OA score and m-F_1 score. BMNet obtains gains of 1.7% and 1.0% in the m-F_1 score and OA scorecompared with HRNet (baseline). However, the segmentation results for comparison methods trained on the Vaihingen dataset are not as good as those on the Potsdam dataset. There are four main reasons leading to the above occurrence. First, the Potsdam dataset has fewer training samples and more testing samples than the Vaihingen dataset. Second, the size of images in the Potsdam dataset is approximately 1/7 of that in the Vaihingen dataset. Third, images in the Potsdam dataset all have the blue channel, which is missing from the images in the Vaihingen dataset. Finally, images in the Potsdam dataset have a smaller ground sampling distance (GSD) than that in the Vaihingen dataset.

Interestingly, the F_1 scores of the proposed BMNet in large-scale objects, such as impervious surfaces and buildings, are slightly higher than that of HRNet-OCR, which also introduces contextual dependencies. Therefore, compared with HRNet, the improvements in large-scale objects prove that the introduction of long-range contextual dependencies can refine the features of large-scale objects. Attributable to the additional additions of scale variations and channel dependencies in the SAM, the BMNet can be better at refining the representation of features in large-scale objects than HRNet-OCR. Besides, BMNet has significantly higher F_1 score than HRNet-OCR in small-scale objects, such as low vegetation, trees, and cars. The significant improvements of BMNet in small-scale objects provide evidence for the fact that the BGM can guide networks to focus on the boundaries and refine the representation of the details of an object, which is an ability the HRNet-OCR does not have.

Fig. 12 presents some visual segmentations on the Vaihingen dataset. Note that there are different categories of objects with extremely similar features, which will result in critical misclassifications. In the first rows of Fig. 12, buildings have a similar color to low vegetation due to the lack of blue channel. Besides, the shadow makes it difficult to distinguish between the low vegetation and the impervious surfaces in the second rows of Fig. 12. Not surprisingly, none of the comparison methods can accurately classify these objects in Fig. 12(c)–(g). After fusing similar features with channel similarity dependencies, BMNet can amplify the differences

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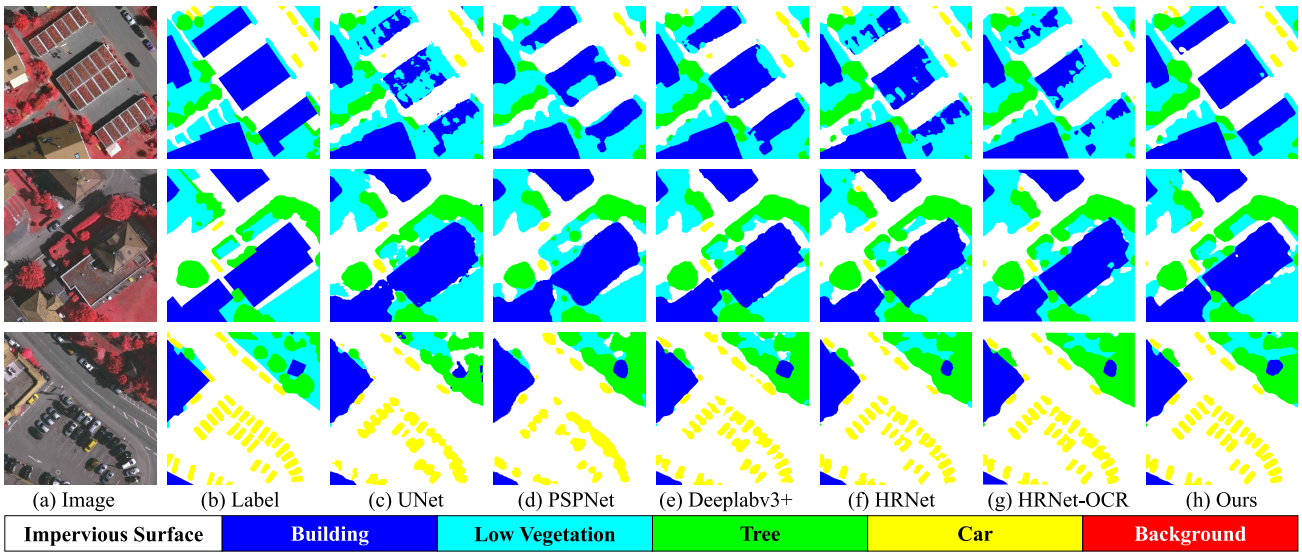


Fig. 12. Comparison of segmentation results between our method and prevalent methods on the Vaihingen test set. (a) Image. (b) Label. (c) UNet. (d) PSPNet. (e) Deeplabv3+. (f) HRNet. (g) HRNet-OCR. (h) Ours.

between features to obtain more discriminative features and accurately distinguish these objects with similar features in Fig. 12(h). To demonstrate the refinement of details of small-scale objects, the last rows of Fig. 12 are displayed. As shown in Fig. 12(c)–(g), the boundaries of cars predicted by comparison methods are incomplete or interconnected, which means that none of the comparison methods can obtain accurate boundary information of small-scale objects. Guided by the boundary attention generated by the BGM, BMNet obtains better segmentation results and successfully solved this problem in Fig. 12(h). Therefore, our method can further optimize the discriminability and the boundary details of objects.

V. Conclusion

In this article, we propose the BMNet that recovers accurate boundaries with manifold information to produce precise segmentation of remote sensing images. To eliminate the influence of the scale variation and the noise disturbance, the SAM is proposed to enhance the discriminability of features by constructing multiscale contextual dependency. In addition, we propose BGM to guide more precise segmentation near the boundary with the boundary information, which is mined from the manifold structure of multiscale features. The proposed BMNet is evaluated on the Potsdam dataset and the Vaihingen dataset, and it obtains better performance in segmentation than prevalent methods, which shows the effectiveness of the BMNet in precise predictions of object details. In the future, we will combine the boundary constraint term of the loss function to learn the boundary information in a supervised manner and optimize the network structure through network architecture search (NAS) methods to improve the representation of features.

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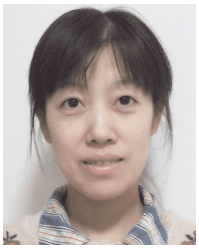
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